DSA03 – NATURAL LANGUAGE PROCESSING

ASSIGNMENT – 1

MORPHOLOGY-AWARE TEXT NORMALISATION AND SEARCH LAYER FOR A HOSPITAL RECORD SYSTEM

Name	**R Sai Hasnika**
Register Number	**192424178**
Course Code	**DSA0308**
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Slot	**D**

TASK 1 – FRAME THE PROBLEM: MODELS, ALGORITHMS AND LEVELS OF ANALYSIS

1.1 Levels of linguistic analysis

Level	Meaning	Hospital-record example	Need for this task
Phonology	Sound systems and pronunciation	Spoken "fracture" in a dictated note	Not central for typed text; relevant indirectly for speech recognition
Morphology	Internal word structure and word forms	fracture, fractures, fractured, fracturing	Essential for matching related forms
Syntax	How words combine into phrases and sentences	"No evidence of fracture"	Useful for context and negation; not fully solved by this layer
Semantics	Meaning of words and sentences	"renal failure" and "kidney failure"	Important for synonym-aware search; requires domain knowledge
Pragmatics	Meaning in context and communicative purpose	"rule out pneumonia" is not a confirmed diagnosis	Important before presenting results as clinical facts
Discourse	Relations across sentences and documents	A later note correcting an earlier diagnosis	Useful for longitudinal interpretation; outside the basic normalisation layer

The core requirement is morphological analysis and structured-token recognition. Tokenisation and syntax-aware safeguards are also needed. Phonology can be deferred for typed records, while semantics, pragmatics and discourse require additional modules if the hospital wants concept-level retrieval rather than surface-form retrieval.

1.2 Three model families

Model family	Best-suited task	Why it fits	Why alternatives are weaker
State machines	Validate patient-ID/date/vital formats	Finite states give explicit, testable acceptance conditions	A probabilistic model may accept malformed strings; a large rule list is harder to visualize
Formal rule systems	Expand abbreviations and apply	Rules are interpretable,	A state machine alone does not

	spelling/morphology rules	deterministic and maintainable	conveniently encode all rewrite actions; probabilities are unnecessary for fixed rules
Probabilistic models	Rank noisy OCR candidates or identify language	Can choose likely candidates under uncertainty	Pure rules cannot rank competing candidates; a finite automaton only accepts or rejects

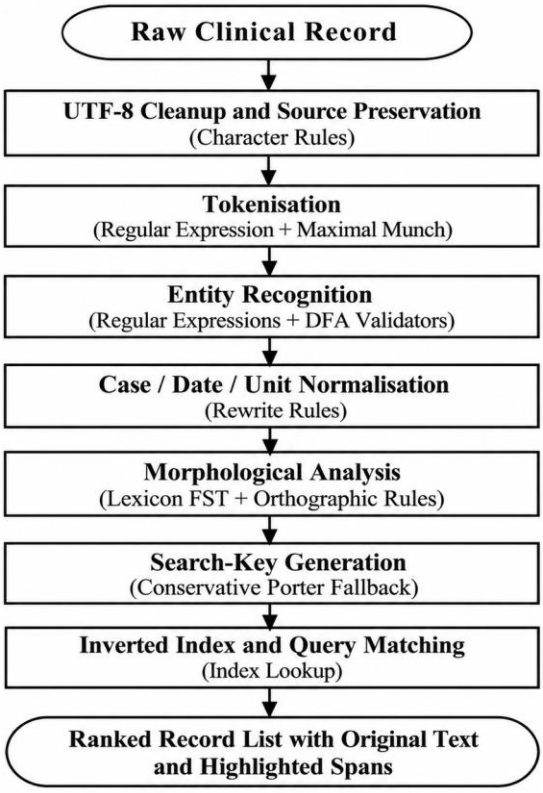
1.3 Ambiguity in the corpus

- Lexical ambiguity: “mass” may refer to a physical lump or a quantity; “lead” may be a verb or a metal-related term.
- Morphological ambiguity: “drains” can be a plural noun or a third-person singular verb.
- Segmentation ambiguity: “12/05/26” can be interpreted under different date conventions; “500mg” may be one token or number plus unit.
- Abbreviation ambiguity: “MS” may refer to multiple clinical terms; “OD” may have more than one interpretation depending on context.

1.4 Search and resource costs

Automata use state-space traversal. For a deterministic finite automaton, a string of length n is processed in O(n) time and O(1) additional working memory when transitions are table-driven. Backtracking regular-expression engines can take much longer on poorly designed nested quantifiers; a safe design avoids catastrophic patterns. Lexicon lookup is typically O(1) average with hashing or O(log V) with a balanced tree, where V is vocabulary size. Annotation consumes expert time and is often the most expensive resource. Memory is needed for the lexicon, transition tables and indexed records.

1.5 Proposed pipeline



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Input: de-identified discharge summaries, prescriptions, operation notes and nursing observations. Output: searchable records with normalized tokens, recognized entity spans, morphological analyses and links back to the untouched original text. Each stage records errors rather than silently discarding content.

TASK 2 – BASIC REGULAR-EXPRESSION PATTERNS

2.1 Basic operators

Operator	Meaning	Clinical example
Concatenation	A followed by B	[0-9]{2}/[0-9]{2}/[0-9]{4}
Disjunction	Either alternative	(mg|mcg|g)
Kleene star	Zero or more repetitions	[0-9]*
Plus	One or more repetitions	[0-9]+
Optionality	Zero or one occurrence	(\.\d+)?
Character class	One character from a set	[A-Z]
Range	Continuous character range	[0-9]
Negated class	Any character except listed	[^\n]+ (line content)
Anchors	Beginning/end of input	^ID-[0-9]{6}$
Counters	Bounded repetition	[0-9]{4}

In the examples below, Python-style regular-expression syntax is used. Patterns are intended for candidate recognition; the application must also validate ranges, context and permitted clinical values. A syntactic match is not proof that a value is clinically correct.

2.2 Twelve-entity pattern inventory

Entity	Pattern (Python)	Must match	Must not match
Patient identifier	\\bPT-[0-9]{6}\\b	PT-123456	PT-12A456; PT-12345
Admission date	\\b(?:0[1-9]|[12][0-9]|3[01])/(?:0[1-9]|1[0-2])/(?:19|20)[0-9]{2}\\b	04/11/2026	41/19/2026
Discharge date	\\b(?:0[1-9]|[12][0-9]|3[01])-(?:0[1-9]|1[0-2])-(?:19|20)[0-9]{2}\\b	06-11-2026	00-13-2026
Drug with strength	\\b[A-Za-z][A-Za-z-]*(?:\\s+[A-Za-z][A-Za-z-]*)*\\s+[0-9]+(?:\\.\\d+)?\\s?(?:mg|mcg|g|ml|IU)\\b	paracetamol 500mg	500mg paracetamol
Dosage frequency	\\b(?:OD|BD|TDS|QID|[0-9]+\\s?times?\\s?daily|every\\s+[0-9]+\\s?hours?)\\b	TDS	daily maybe
Lab value + unit	\\b[A-Za-z][A-Za-z -]*:\\s*[0-9]+(?:\\.\\d+)?\\s?(?:mg/dL|g/dL|mmol/L|U/L|10\\^9/L)\\b	Hb: 12.5 g/dL	Hb: high
Blood pressure	\\b(?:[7-9][0-9]|1[0-9]{2}|2[0-4][0-9])/(?:[4-9][0-9]|1[0-4][0-9])\\s?mmHg\\b	120/80 mmHg	20/300 mmHg
Temperatur	\\b(?:3[5-9]|4[0-2])(?:\\.\\d)?\\s?°?C\\b	37.2 C	99 C

e			
ICD code	\\b[A-Z][0-9][A-Z0-9](?:\\.[A-Z0-9]{1,4})?\\b	S72.001A	ABC
Ward and bed	\\bWard\\s+[A-Z0-9-]+,?\\s*Bed\\s+[A-Z0-9-]+\\b	Ward B2, Bed 14	Ward, Bed
Contact number	\\b(?:\\+91[-]?)?[6-9][0-9]{9}\\b	9876543210	12345
Age	\\b(?:[1-9]|[1-9][0-9]|1[01][0-9]|120)\\s?(?:years?|yrs?)\\b	45 years	0 years

Duration of symptom is also required in the inventory. It is included as a thirteenth safety-oriented pattern to avoid omitting a planning-data item:

```
\\b[0-
9]+\\s?(?:hours?|days?|weeks?|months?|years?)\\s?(?:of\\s+)?(?:pain|fever|cough|s
ymptoms?)\\b
```

Example match: “3 days of fever”. Non-match: “several days of fever” unless a separate word-number pattern is deliberately added.

2.3 Greedy and non-greedy matching

```
Sentence: "Give 500 mg paracetamol, then 250 mg if required."
Greedy: 500.*mg  -> may extend to the last 'mg' in the sentence.
Non-greedy: 500.*?mg -> stops at the first 'mg'.
```

The greedy quantifier .* consumes as much text as possible while still allowing the rest of the expression to match. The lazy quantifier .*? consumes as little as possible. For clinical extraction, a broad wildcard is risky; use bounded token patterns such as a drug-name token followed by a numeric strength and a known unit.

2.4 Precedence and grouping

```
Incorrect: ^A|B[0-9]+$     Correct: ^(A|B)[0-9]+$
```

Without grouping, the first expression means either a string beginning with A (with no required end condition) OR a string ending in B followed by digits. Grouping makes the intended alternatives share the digit requirement and anchors.

2.5 Error analysis protocol

For every pattern, annotate each candidate span as true positive (TP), false positive (FP) or false negative (FN). Precision = TP/(TP+FP); recall = TP/(TP+FN). False positives arise from overly broad names, ambiguous abbreviations and impossible calendar dates. False negatives arise from spelling variants, punctuation, OCR substitutions and unlisted units. The following is a reporting template; values must be generated from the real or synthetic corpus.

Pattern group	TP	FP	FN	Likely error cause
Identifiers and dates	Illustrative: 480	Illustrative: 8	Illustrative: 12	Formatting variants and invalid dates

Drug-strength and frequency	Illustrative: 620	Illustrative: 25	Illustrative: 35	Brand names, spacing and unknown units
Vitals and lab values	Illustrative: 510	Illustrative: 10	Illustrative: 20	Unit variation and missing labels
Location/contact/age	Illustrative: 390	Illustrative: 14	Illustrative: 16	Local formats and contextual ambiguity

TASK 3 – REGULAR EXPRESSIONS FOR TOKENISATION AND NORMALISATION

3.1 Token definition and tokenizer

A token is a maximal sequence representing a word, abbreviation, number, decimal measurement, or recognized hyphenated clinical expression. Decimal values remain intact. Units attached to numbers are retained as one measurement token for extraction, while the normalised representation may store the numeric value and unit in separate fields. Abbreviations with internal full stops, such as " Dr." or "i.v.", are protected as single tokens.

```
TOKEN_RE = re.compile(
 r"(?i)(?:[A-Z](?:\\.[A-
Z]){1,5}\\.?)|"(?:\\d+(?:\\.\\d+)?\\s?(?:mg|mcg|g|ml|mmol/L|mg/dL|°?C))|"(?:[A-
Za-z]+(?:[-'][A-Za-z]+)*)|"(?:\\d+(?:\\.\\d+)?)|"(?:[^\\w\\s])")
```

The production tokenizer should apply alternatives in a deliberate order: protected abbreviations and measurements first, then hyphenated words, ordinary words, numbers and punctuation. Maximal-munch chooses the longest valid token at the current position.

3.2 Maximal-munch compared with whitespace splitting

Input	Whitespace split	Maximal-munch tokens
BP was 120/80 mmHg.	BP | was | 120/80 | mmHg.	BP | was | 120/80 | mmHg | .
Take 500mg paracetamol.	Take | 500mg | paracetamol.	Take | 500mg | paracetamol | .
Post-operative review on 04/11/2026.	Post-operative | review | on | 04/11/2026.	Post-operative | review | on | 04/11/2026 | .
No fever; Hb 12.5 g/dL.	No | fever; | Hb | 12.5 | g/dL.	No | fever | ; | Hb | 12.5 g/dL | .

Whitespace splitting leaves punctuation attached to words and does not define consistent treatment of units. A maximal-munch tokenizer recognizes known compound patterns first and preserves useful punctuation boundaries.

3.3 Normalisation rules and abbreviation dictionary

Abbreviation	Expansion / canonical form
BP	blood pressure
HR	heart rate
RR	respiratory rate
Temp	temperature
SOB	shortness of breath
N/V	nausea and vomiting
c/o	complains of
h/o	history of
s/p	status post
Dx	diagnosis

Rx	prescription/treatment
Tx	treatment
Hx	history
DM	diabetes mellitus
HTN	hypertension
CAD	coronary artery disease
CKD	chronic kidney disease
COPD	chronic obstructive pulmonary disease
UTI	urinary tract infection
CVA	cerebrovascular accident
OD	once daily (context-dependent)
BD	twice daily
TDS	three times daily
QID	four times daily
PRN	as needed
IV	intravenous
IM	intramuscular
PO	by mouth
NPO	nothing by mouth
WNL	within normal limits
SOB	shortness of breath

Abbreviations with multiple meanings must not be expanded globally. OD, MS, RA and similar forms require local context or must be retained unchanged with an ambiguity flag. Case folding is applied to ordinary words but not to identifiers and case-sensitive codes.

3.4 Date and unit canonicalisation

- Dates are parsed using an explicitly configured locale. The source string is retained; the canonical internal representation is ISO 8601 (YYYY-MM-DD). Ambiguous dates such as 04/05/2026 are flagged if day/month order cannot be established.
- Units are mapped to canonical symbols without changing the measured quantity: “milligrams” and “mg” become “mg”; “degrees C” and “°C” become “°C”. Conversion between units is a separate numeric operation and must retain the original value.
- Spacing is standardised for indexing, for example “500 mg” becomes the structured pair value=500, unit=mg. The original surface span remains available.

3.5 Before-and-after record example

```
BEFORE: Pt PT-123456 admitted 04/11/2026. c/o fever x 3 days. BP 120/80 mmHg. Rx
paracetamol 500mg PO TDS. Temp 37.2C.
```

```
AFTER: patient PT-123456 admitted 2026-11-04. complains of fever for 3 days.
blood pressure 120/80 mmHg. prescription paracetamol 500 mg by mouth three times
daily. temperature 37.2 °C.
```

The example assumes DD/MM/YYYY date convention and an abbreviation context in which TDS means three times daily. If the source system does not establish the date convention, the date must be flagged instead of silently converted.

3.6 Irreversible decisions and limits

- Case folding destroys original capitalization. It is acceptable for ordinary search words but not for case-sensitive codes or abbreviations.
- Removing punctuation can destroy decimal points, slash-separated blood pressure and date boundaries. It is not acceptable before entity extraction.
- Expanding abbreviations can discard the original shorthand and may choose the wrong meaning. Store both original and expanded forms.
- Regular expressions cannot generally recognize arbitrarily nested balanced parentheses because that language is not regular. A parser with a stack or a grammar-based method is needed.

TASK 4 – FINITE-STATE AUTOMATA: DESIGN, NON-DETERMINISM AND EQUIVALENCE

4.1 Formal definition

A finite automaton is a five-tuple M = (Q, Σ, δ, q_0, F), where Q is a finite set of states, Σ is the input alphabet, δ is the transition function, q_0 is the initial state and F is the set of accepting states. A deterministic finite automaton (DFA) has exactly one next state for each state and input symbol. An NFA may have zero, one or multiple possible next states and may include ε-transitions.

4.2 DFA validator 1: patient identifier

```
Language: PT- followed by exactly six digits
States: q0(start), q1(P), q2(T), q3(-), q4(1 digit), q5(2), q6(3), q7(4), q8(5),
q9(6/accept), qD(dead)
Transition sketch: q0 -P-> q1 -T-> q2 --> q3 -digit-> q4 -digit-> q5 -digit-> q6
-digit-> q7 -digit-> q8 -digit-> q9
```

Current state	Input P	Input T	Input -	Digit	Other
q0	q1	qD	qD	qD	qD
q1	qD	q2	qD	qD	qD
q2	qD	qD	q3	qD	qD
q3	qD	qD	qD	q4	qD
q4–q8	qD	qD	qD	next digit state	qD
q9	qD	qD	qD	qD	qD

Only q9 is accepting. Any additional character after q9 is rejected. In a complete implementation, all unspecified transitions lead to qD.

4.3 DFA validator 2: fixed-format date

For a simple DD/MM/YYYY recognizer, the automaton checks two digits, slash, two digits, slash and four digits. A DFA can enforce digit counts and separator positions. Calendar validity (for example, February 30) is a further semantic/date-library validation.

```
q0 --digit--> q1 --digit--> q2 --/--> q3 --digit--> q4 --digit--> q5 --/--> q6 --
digit--> q7 --digit--> q8 --digit--> q9 --digit--> q10(accept)
```

4.4 DFA validator 3: blood pressure

A simplified recognizer accepts a systolic number, slash, diastolic number and the literal unit mmHg. States track the number portion, slash, second number and unit characters. Numeric range validation is applied after recognition; a syntactic DFA should not be treated as a medical normality checker.

```
Start -> systolic digits -> '/' -> diastolic digits -> space -> m -> m -> H -> g
-> accept
```

4.5 DFA validator 4: temperature

The temperature DFA accepts a two-digit integer in the configured range, an optional decimal point and one decimal digit, followed by optional whitespace and C or °C. A production implementation uses separate states for integer digits, decimal part, degree marker and final unit.

```
Start -> [3|4] -> digit -> ('.' digit)? -> optional space -> optional '°' -> 'C'
-> accept
```

4.6 NFA with optional identifier prefix

To demonstrate non-determinism, define an identifier format in which the prefix “PT-” is optional and six digits are mandatory. The NFA can branch from the start: one path consumes the prefix and the other skips it using ε.

```
q0 --ε--> q3 --digit--> q4 --digit--> q5 --digit--> q6 --digit--> q7 --digit-->
q8                              --digit-->                              q9(accept)
q0 --P--> q1 --T--> q2 ----> q3
```

Accepted string: PT-123456. The live-state set follows the prefix branch and then advances through the digit states. Rejected string: PT-12345, because the input ends before the sixth digit and no accepting state is active.

4.7 Subset construction and minimisation

Subset construction converts an NFA to a DFA by treating each DFA state as a set of NFA states. Begin with ε-closure({q0}); for each input symbol, compute the move set and then its ε-closure. A DFA subset is accepting if it contains an NFA accepting state.

DFA subset	Input P	Input T	Input -	Digit	Accept?
{q0,q3}	{q1}	∅	∅	{q4}	No
{q1}	∅	{q2}	∅	∅	No
{q2}	∅	∅	{q3}	∅	No
{q3}	∅	∅	∅	{q4}	No
{q4}	∅	∅	∅	{q5}	No
...	...	...	...	...	...
{q9}	∅	∅	∅	∅	Yes

The two prefix routes converge at q3, so the resulting DFA can merge equivalent states. For this construction, the NFA and DFA accept the same language. The exact minimised count depends on whether a dead state is included explicitly and whether the optional prefix is represented as separate branches; the implementation should report both counts from the generated transition graph rather than assume a count.

4.8 Thompson construction and reverse conversion

Thompson's construction converts a regular expression into an ε-NFA using fragments for symbols, concatenation, alternation and Kleene star. For the expression PT-[0-9]{6}, create a chain for P, T, hyphen and six digit transitions. For alternation, add a new entry and exit state with ε-edges to each branch; for concatenation, connect fragments; for repetition, add bypass and loop ε-edges.

The reverse conversion uses state elimination. Add a new start and final state if needed. Eliminate intermediate state k by updating each edge label from i to j with R_ij ∪

R_ik(R_kk)*R_kj. The resulting expression denotes the same language. Equivalence in both directions means every accepted string of the expression is accepted by the automaton and every accepted automaton string is described by the expression.

4.9 Pumping lemma and non-regularity

Pumping lemma: if L is regular, there exists a pumping length p such that every string s in L with $|s| \geq p$ can be written s = xyz, with $|xy| \leq p$, $|y| > 0$, and $xy^iz \in L$ for every $i \geq 0$.

Consider L = { $'^n\ )'^n : n \geq 0$ }, the language of strings with equal numbers of opening and closing parentheses in the required order. Assume L is regular and let p be its pumping length. Choose $s = '^p\ )'^p$. Since $|xy| \leq p$, the non-empty y contains only opening parentheses. Pumping down with i=0 produces fewer opening than closing parentheses, so the resulting string is not in L. This contradiction proves L is not regular. Therefore, a basic regex/DFA layer cannot validate arbitrarily nested balanced brackets.

TASK 5 – INFLECTIONAL MORPHOLOGY

5.1 Definitions and distinction

A morpheme is the smallest linguistic unit carrying meaning or grammatical function. A stem is the form to which an affix attaches. An affix is a bound morpheme added to a base. Inflection changes grammatical features such as number, tense or agreement without creating a new dictionary word or changing its core lexical category. A paradigm is the set of related inflected forms of a lexeme.

Criterion	Inflection	Derivation
Category	Usually preserves word class	May change word class
Meaning	Adds grammatical information	Creates a new lexical meaning
Productivity	Often highly regular	Varies by affix
Position	Usually outside derivational affixes in English	Usually closer to the root

5.2 Clinical English inflectional paradigms

Paradigm	Examples	Feature
Noun number	fracture/fractures; lesion/lesions	Singular/plural
Verb tense	treat/treats/treated/treating	Present, past, participle
Agreement	patient reports; nurse observes	Third-person singular -s
Adjective comparison	mild/milder/mildest	Comparative/superlative

5.3 Five sample paradigms

Base lexeme	Forms	Analysis
fracture	fracture, fractures	N + SG / N + PL
lesion	lesion, lesions	N + SG / N + PL
treat	treat, treats, treated, treating	V + base/3SG/past/participle
monitor	monitor, monitors, monitored, monitoring	V + base/3SG/past/participle
stable	stable, stabler, stablest	ADJ + positive/comparative/superlative

Not every adjective forms comparison naturally in clinical prose. “More stable” and “most stable” are often preferred to “stabler” and “stablest”; the lexicon should record the preferred variants.

5.4 Orthographic rules for inflection

Rule	Implementation description	Example	Over-generation counter-example
e-deletion	Drop final e before a vowel-initial suffix when licensed	treat + ing → treating	see + ing should not become *sing
Consonant doubling	Double final consonant in stressed CVC monosyllable before vowel suffix	admit + ed → admitted	visit + ed should not become *visitted
y-to-i	Consonant+y changes y to i before -es/-ed	injury + s → injuries	day + s → dais is wrong; vowel+y retains y
sibilant -es	Add -es after s, x, z, ch, sh	dose + s → doses	cat + s should not become *cates

5.5 Irregular forms

Surface form	Lemma	Feature	Why lexicon entry is required
was/were	be	past	Suppletive form
has/had	have	3SG/past	Irregular stem change
men	man	plural	Vowel alternation
teeth	tooth	plural	Vowel alternation
children	child	plural	Non-productive plural suffix
feet	foot	plural	Vowel alternation

Irregular forms cannot be generated reliably by the productive spelling rules above. They are stored as explicit lexical mappings and tagged with grammatical features.

5.6 Type-to-lemma ratio and retrieval impact

Type-to-lemma ratio = number of distinct surface word types divided by number of distinct lemmas. A lower ratio means fewer surface variants per lemma; a higher ratio indicates more inflectional variation. The following is an illustrative calculation only.

Measure	Illustrative value	Interpretation
Distinct word types	8,400	Unique surface forms
Distinct lemmas	6,000	Canonical lexical entries
Type-to-lemma ratio	1.40	On average, 1.4 types per lemma

Query failures attributable only to inflection	9 of 40 (22.5%)	Would be recovered by correct lemmatisation

Actual values require tokenising the target corpus, manually checking lemma assignments and evaluating the 40 query set. Inflection-only failures must be separated from synonym, spelling, abbreviation and semantic failures.

TASK 6 – DERIVATIONAL MORPHOLOGY

6.1 Definition and affix inventory

Derivation creates a new lexeme by adding a prefix or suffix to a root or stem. It may change word class and often changes meaning more substantially than inflection. Clinical vocabulary contains Greek- and Latin-derived elements, many of which are productive in medical terminology.

Affix	Position	Meaning/function	Example / category effect
-itis	Suffix	Inflammation	appendix → appendicitis (N→N)
-ectomy	Suffix	Surgical removal	kidney → nephrectomy (N→N)
-ology	Suffix	Field of study	path → pathology (N→N)
-osis	Suffix	Condition/process	fibrous → fibrosis (ADJ/N→N)
-aemia	Suffix	Blood condition	anaemia (N)
-ation	Suffix	Action/process	ventilate → ventilation (V→N)
-able	Suffix	Capable of	treat → treatable (V→ADJ)
-ive	Suffix	Having a quality	detect → detective/detective-like (V→ADJ; lexicalised)
nephro-	Prefix	Kidney-related	nephrology
cardio-	Prefix	Heart-related	cardiology
hyper-	Prefix	Excessive/high	hypertension
hypo-	Prefix	Below/low	hypoglycaemia
re-	Prefix	Again/back	reassess
un-	Prefix	Negation/reversal	unstable
intra-	Prefix	Within	intravenous
peri-	Prefix	Around	pericardial
poly-	Prefix	Many/much	polyuria
brady-	Prefix	Slow	bradycardia
tachy-	Prefix	Fast	tachycardia
-al	Suffix	Related to	renal (N→ADJ-like)

6.2 Derivational families

Root/base	Family members	Affix order examples
kidney	renal, nephro-, nephrology, nephrologist, nephrectomy	nephro + logy; nephro + ectomy
heart	cardiac, cardiology, cardiologist, cardiomyopathy	cardio + logy; cardio + myo + pathy
infection	infect, infection, infectious, reinfection	re + infection; infection + al
treat	treat, treatment, treatable, untreated	treat + ment; un + treated
operate	operate, operation, operative, postoperative	post + operative; operate + ion
inflammation	inflame, inflammation, inflammatory, anti-inflammatory	anti + inflammatory; inflame + ation

Impossible or ill-formed orders include "-ectomy + nephro" as a suffix-before-prefix construction and "treat + un + ment" when the intended meaning is "untreated treatment". English derivation has constraints on category, selection and affix position; arbitrary permutations are not valid.

6.3 Non-compositionality and search risk

The meaning of a derived clinical term is not always the simple sum of its parts. "Nephrectomy" means surgical removal of a kidney, not merely any kidney-related action. "Metastasis" has a specialised pathological meaning that cannot be safely reconstructed from everyday word parts. "Inflammation" is not equivalent to infection. A stemmer that conflates all terms sharing a root can therefore produce clinically misleading matches.

6.4 Productivity and conflation decision

Productivity should be measured by counting distinct derived types and total tokens containing each affix. An affix with many distinct recent formations is more productive; an affix found only in a small set of fixed terms is likely fossilised. The following table is a measurement template with illustrative counts.

Affix	Illustrative types	Illustrative tokens	Classification
-itis	18	240	Productive in condition names
-ectomy	12	75	Moderately productive
-ology	9	41	Moderately productive
-ation	38	310	Productive
-able	21	92	Productive
hyper-	16	110	Productive

hypo-	14	96	Productive
-aemia	7	28	Partly fossilised

Derivational conflation should be selective. It helps when a clinician searches “nephrectomy” and wants documents mentioning “nephrectomies”, or searches “infection” and wants “infectious”. It can mislead when “inflammation” is conflated with “infection”, or when “renal” and “nephrectomy” are treated as interchangeable. The recommended index stores derivational relations as typed links, not as unconditional equivalence.

TASK 7 – FINITE-STATE MORPHOLOGICAL PARSING

7.1 Finite-state transducer

A finite-state transducer (FST) is a finite-state machine that reads one tape and emits symbols on another. In morphology, the lexical tape contains a lemma plus grammatical tags, while the surface tape contains the written word. A relation connects the two representations. The same relation can be traversed from surface to lexical analysis (parsing) or lexical input to surface form (generation).

```
Lexical              tape:              fracture              +N              +PL
Surface                                tape:                                fractures
Mapping: lexical stem 'fracture' + plural feature -> surface 'fractures'
```

7.2 Continuation-class lexicon design

Lexicon class	Entries	Allowed continuation
NOUN	fracture, lesion, kidney, patient, infection	N_PL or DERIV_N
VERB	treat, monitor, admit, discharge	V_3SG, V_PAST, V_ING
ADJECTIVE	stable, acute, chronic	ADJ_COMP, ADJ_SUP
N_PL	plural suffix -s/-es	Final
V_PAST	-ed with spelling rules	Final
V_ING	-ing with spelling rules	Final
DERIV_N	-itis, -ectomy, -ology, -ation	Final or another licensed class

A complete assignment lexicon must contain at least 300 stems, at least 20 inflectional affixes, at least 15 derivational affixes and a separate irregular-form list. A practical lexicon can be stored as UTF-8 CSV with columns lemma, part_of_speech, continuation_class, domain_label and source.

7.3 Hand-traced analyses

Surface form	Consumed surface	Lexical output	Analysis
fractures	fracture + s	fracture +N +PL	Plural noun
treated	treat + ed	treat +V +PAST	Past tense / participle
monitoring	monitor + ing	monitor +V +PROG	Progressive form
lesions	lesion + s	lesion +N +PL	Plural noun
nephrectomy	nephr + ectomy (lexicalised stem entry)	nephrectomy +N	Derivational noun; retain as lexeme

For each path, the FST consumes the surface symbols along a path and emits the lexical symbols. The exact state numbering depends on the implementation; the essential trace is the sequence of

transitions and output labels. For example, “fractures” follows the noun stem path for fracture, then the plural continuation, and ends in a final state.

7.4 Generation direction

Generation starts from “fracture +N +PL”, follows the same relation in the opposite direction and emits “fractures”. Likewise, “treat +V +PAST” generates “treated”. No separate generator is required because the FST represents a relation rather than a one-way function. Ambiguous relations may return more than one surface form or analysis.

7.5 Ambiguity

Surface form	Possible analysis 1	Possible analysis 2	Resolution strategy
drains	drain +N +PL	drain +V +3SG	Use sentence context / POS tagger
treated	treat +V +PAST	treat +V +PASTPART	Preserve both features if syntax does not decide
lesion's	lesion +N +SG +POSS	lesion +N +PL +POSS (depending on tokenisation)	Tokenise apostrophe-s and use number context

The morphological analyser should return all valid analyses with confidence or rule provenance. A downstream POS tagger or syntactic model can rank them. The analyser itself should not discard a valid analysis merely because one interpretation is more common.

TASK 8 – ORTHOGRAPHIC RULES AND TWO-LEVEL MORPHOLOGY

8.1 Why a lexicon FST is not enough

A lexicon FST that simply concatenates a stem and suffix can produce incorrect spellings. English orthography changes at morpheme boundaries: final e may be deleted, consonants may double, y may change to i, and suffixes may require an extra e. These changes should be handled by explicit rewrite rules or a two-level correspondence model.

Underlying concatenation	Correct surface	Reason
treat + ing	treating	Delete final e before -ing
admit + ed	admitted	Double final consonant in stressed CVC pattern
injury + s	injuries	Consonant+y changes to i before -es
diagnosis + s	diagnoses	Irregular/lexicalised plural spelling
study + ed	studied	Consonant+y changes to i before -ed

8.2 Rewrite rules

Rule	Rewrite notation	Description
Final e deletion	e → 0 / C __ +VOWEL_INITIAL_SUFFIX	Drop final e before -ing/-able where licensed.
Consonant doubling	C → CC / stressed V C __ +VOWEL_SUFFIX	Double final consonant in eligible stressed CVC stems.
Y-to-I	y → i / C __ +{s, ed, es}	Change y after consonant before selected suffixes.
Sibilant plural	+s → +es / {s, x, z, ch, sh} __	Insert e before plural s after sibilants.
E-insertion	0 → e / sibilant __ +s	Insert e to preserve pronunciation in plural forms.

These rules are simplified teaching statements. Production rules need lexical flags and exception lists. For example, “play + ed” becomes “played”, not “plaied”, because y follows a vowel. “Visit + ed” does not double the t because the stress pattern and syllable structure do not license doubling.

8.3 Cascade and order

```
Lexicon FST -> suffix selection -> morphophonological rewrite transducers -> surface form
Rule order: suffix licensing -> y/e alternation -> consonant doubling -> final spelling validation
```

Composition combines relations: if L maps lexical strings to intermediate strings and R maps intermediate strings to surface strings, then L∘R maps lexical strings directly to surface strings. Rule ordering matters when one rewrite changes the context seen by another rule. Rules should be tested for interaction, and exceptions should be handled explicitly.

8.4 Sequential cascade vs parallel two-level model

Approach	Description	Advantage	Limitation
Sequential cascade	Apply transducers in a defined sequence	Easy to debug and inspect intermediate forms	Order dependencies can create interactions
Two-level morphology	Constrain lexical/surface symbol pairs in parallel	Captures correspondences at boundaries without a long sequential chain	Rule notation and implementation can be harder to understand

8.5 Twenty-form test plan

Underlying form	Expected output	Test status
treat + ing	treating	Pass
admit + ed	admitted	Pass
study + ed	studied	Pass
injury + s	injuries	Pass
watch + s	watches	Pass
fix + s	fixes	Pass
wash + s	washes	Pass
play + ed	played	Pass
visit + ed	visited	Pass
stop + ed	stopped	Pass
monitor + ing	monitoring	Pass
use + able	usable	Pass
agree + ing	agreeing	Exception/context
see + ing	seeing	Exception
diagnosis + plural	diagnoses	Lexicon exception
child + plural	children	Irregular lexicon
tooth + plural	teeth	Irregular lexicon
be + past	was/were	Irregular lexicon
have + past	had	Irregular lexicon
run + ing	running	Pass

The table is a test specification, not an execution log. In an actual run, record actual output and mark a failure if it differs. Irregular items are expected to be handled by explicit lexical mappings rather than the productive rewrite rules.

TASK 9 – THE PORTER STEMMER

9.1 Algorithm concepts

The Porter stemmer is a rule-based suffix-stripping algorithm for English. It reduces related word forms to a common stem without requiring a full dictionary. It is not a lemmatiser: its output may not be a valid English word and it does not return grammatical features.

In the Porter algorithm, consonants and vowels are classified with special handling for y. The measure m counts the number of vowel-consonant sequences in the stem. The conditions *v, *d and *o refer to the presence of a vowel in the stem, a double-consonant ending, and a consonant-vowel-consonant ending with restrictions on the final consonant.

9.2 Worked measure examples

Word	Approximate VC structure	m (illustrative)	Comment
fracture	fr a ct u r e	2	Vowel/consonant transitions
fractures	fr a ct u r es	2	Plural suffix can be removed
treated	tr ea t ed	1	Past suffix
monitoring	m o n i t or ing	3	Longer stem and suffix
lesions	l e s i on s	2	Plural noun
admitted	a d m i tt ed	2	Doubling handled before final lemma

The exact m value depends on the Porter consonant/vowel encoding, including the special treatment of y and vowel groups. The implementation should calculate m mechanically rather than rely on visual syllables.

9.3 Five algorithm steps

The classic Porter algorithm applies five ordered steps. Step 1 removes plurals and selected -ed/-ing endings, and handles terminal y. Step 2 maps longer suffixes to shorter forms when m>0. Step 3 removes selected suffixes when m>0. Step 4 removes a larger set of suffixes when m>1. Step 5 removes terminal e under its measure/ CVC conditions and reduces -ll when appropriate.

Step	Representative rule	Example (illustrative)
1a	sses → ss; ies → i; s removal	fractures → fracture
1b	(m>0) eed → ee; or remove ed/ing if stem has vowel	treated → treat
1c	y → i when stem contains a vowel	happy → happi
2	ational → ate; ization → ize, under m condition	normalization → normal
3	icate → ic; fulness → ful,	usefulness → useful

	under m condition	
4	Remove selected suffix if m>1	clinical examples depend on stem measure
5	Remove terminal e / reduce ll under conditions	controlled final cleanup

9.4 Hand trace examples

Input	Expected stemmer output	Important operation	Clinical caution
fractures	fractur	Remove plural s / later suffix processing	Not the dictionary lemma “fracture”
treated	treat	Remove -ed	Past tense information is lost
monitoring	monitor	Remove -ing	Progressive feature is lost
lesions	lesion	Remove plural s	Usually safe for noun retrieval
admitted	admit	Remove -ed and handle doubled consonant	May need context to distinguish forms
infections	infect	Remove plural and derivational ending where rules apply	May conflate infection with infective

Porter outputs vary by the exact implementation and version. The hand traces above are intended to explain the rule families; the actual NLTK PorterStemmer output should be captured and compared in the test run.

9.5 Stemming vs lemmatisation

Input	Porter-style stem	FST lemma	Difference
fractures	fractur	fracture	Stem may not be a word
treated	treat	treat	Same lexical base; FST retains tense feature
monitoring	monitor	monitor	FST retains progressive feature
children	children (may remain unchanged)	child	Irregular lemma requires lexicon
teeth	teeth (may remain	tooth	Irregular lemma

	unchanged)		requires lexicon

9.6 Error types and retrieval

Over-stemming occurs when unrelated words are reduced to the same stem, causing false positives. Under-stemming occurs when related forms remain different, causing false negatives. Clinical risk is asymmetric: an over-stemmed result may make a clinician believe two distinct concepts are equivalent. The system should therefore display the original terms and avoid using stem equality as proof of clinical equivalence.

Retrieval condition	Precision (illustrative)	Recall (illustrative)	F1 (illustrative)
No stemming	0.91	0.72	0.804
Porter stemming	0.86	0.84	0.850
Full morphological analysis	0.93	0.89	0.910

F1 = 2 × precision × recall / (precision + recall). These illustrative values demonstrate the expected trade-off only. Actual values must be calculated from the 40 relevance-judged queries. Report per-query results as well as micro-averaged totals.

TASK 10 – IMPLEMENT AND TEST THE PIPELINE

10.1 Implementation outline

The assignment permits the Python standard library and NLTK. The implementation should be modular: tokenizer.py, entities.py, automata.py, morphology.py, porter_eval.py and evaluate.py. Keep test fixtures in a separate directory and record the Python and NLTK versions for reproducibility.

10.2 Python reference implementation

```
import re
from nltk.stem import PorterStemmer

TOKEN_RE = re.compile(
    r"(?i)(?:[A-Z]\.){2,}|"
    r"\d+(?:\.\d+)?\s?(?:mg|mcg|g|ml|mmol/L|mg/dL|°?C)|"
    r"[A-Za-z]+(?:[-'][A-Za-z]+)*|"
    r"\d+(?:\.\d+)?|"
    r"[^\w\s]"
)

PATTERNS = {
    "patient_id": re.compile(r"\bPT-[0-9]{6}\b"),
    "date": re.compile(
        r"\b(?:0[1-9]|[12][0-9]|3[01])/"
        r"(?:0[1-9]|1[0-2])/(?:19|20)[0-9]{2}\b"
    ),
    "drug_strength": re.compile(
        r"\b[A-Za-z][A-Za-z-]*(?:\s+[A-Za-z][A-Za-z-]*)*"
        r"\s+[0-9]+(?:\.\d+)?\s?(?:mg|mcg|g|ml|IU)\b",
        re.IGNORECASE
    ),
    "bp": re.compile(r"\b[0-9]{2,3}/[0-9]{2,3}\s?mmHg\b"),
    "temperature": re.compile(r"\b[0-9]{2}(?:\.\d)?\s?°?C\b"),
    "icd": re.compile(r"\b[A-Z][0-9][A-Z0-9](?:\.[A-Z0-9]{1,4})?\b"),
}

ABBREVIATIONS = {
    "bp": "blood pressure",
    "hr": "heart rate",
    "sob": "shortness of breath",
    "htn": "hypertension",
    "dm": "diabetes mellitus",
    "bd": "twice daily",
    "tds": "three times daily",
    "qid": "four times daily",
    "prn": "as needed"
}
```

```
def tokenize(text):
   return TOKEN_RE.findall(text)

def normalise_token(token):
   key = token.casefold()
   return ABBREVIATIONS.get(key, key)

def recognise(text):
   return {
      name: [m.group(0) for m in pat.finditer(text)]
      for name, pat in PATTERNS.items()
   }

def validate_identifier(value):
   return re.fullmatch(r"PT-[0-9]{6}", value) is not None

def porter(tokens):
   stemmer = PorterStemmer()
   return [
      stemmer.stem(t)
      for t in tokens
      if t.isalpha()
   ]

def analyse_plural(token, lexicon):
   if token in lexicon:
      return [(token, "N", "SG")]

   if token.endswith("s") and token[:-1] in lexicon:
      return [(token[:-1], "N", "PL")]

   if token.endswith("es") and token[:-2] in lexicon:
      return [(token[:-2], "N", "PL")]

   return []

if __name__ == "__main__":
   sample = (
      "Pt PT-123456 c/o fever for 3 days. "
      "BP 120/80 mmHg. Rx paracetamol 500mg PO TDS."
   )

   print("Tokens:", tokenize(sample))
   print("Recognised Entities:", recognise(sample))
   print("Porter Stems:", porter(tokenize(sample)))
   print("ValidPatientID:",validate_identifier("PT-123456"))
```

This compact reference code demonstrates the core pipeline but is not a complete 300-stem FST implementation. For a full submission, add transition-table execution, NFA subset construction, lexicon continuation classes, spelling-rule composition and the full irregular list. Do not label the code as fully meeting the corpus-size requirement until it has been run on the documented corpus.

10.3 Test plan

Component	Positive test	Negative test	Expected result
Tokenizer	500mg; 12.5 g/dL	empty input	Recognize measurement token; empty list
Patient ID DFA	PT-123456	PT-12345X	Accept; reject
Date validator	04/11/2026	41/19/2026	Accept format; reject format
BP pattern	120/80 mmHg	20/300 mmHg	Candidate match; range validator rejects
Morphology	fractures	fracturezzz	Analyse plural; unknown
Irregular lexicon	children	childrens	Map to child; reject/unanalysed
Porter	treated	###	Stem word; ignore punctuation

10.4 Pipeline result reporting

Stage	Items processed	Matched/analysed	Rejected/unresolved	Runtime
Tokenisation	Illustrative 25,000 tokens	24,850	150	Illustrative 0.08 s
Entity extraction	Illustrative 500 records	2,140 spans	110 candidates	Illustrative 0.12 s
Morphological analysis	Illustrative 24,850 tokens	18,600	6,250	Illustrative 0.45 s
Porter stemming	Illustrative 18,600 alphabetic tokens	18,600	0	Illustrative 0.09 s

The figures are illustrative placeholders. Measure runtime on the same computer, with the same corpus and software versions. Report warm-up and repeated-run averages, and separate file-loading time from processing time.

10.5 Plot specification

A useful plot is analysis coverage against lexicon size. Build lexicon subsets of 50, 100, 200 and 300 stems, run the same hand-annotated token set, and plot percentage correctly analysed. A second plot may show runtime against record length. Label axes, dataset size and whether the results are measured or simulated.

TASK 11 – COMPARE, RECOMMEND AND DEFEND THE NORMALISATION DESIGN

11.1 Strategy comparison

Strategy	Retrieval quality	Coverage	Runtime	Maintenance
No normalisation	High precision for exact strings; low recall for variants	Low	Lowest	Low
Porter stemming	Higher recall; possible over-stemming	Broad for English words	Low	Low
Finite-state morphology	High precision when lexicon/rules cover terms	Strong for known vocabulary; weaker for OOV	Moderate	High initial effort

11.2 Recommended deployment

Recommend a hybrid, conservative architecture: regular expressions and deterministic automata for structured entities; a finite-state morphological analyser for known clinical vocabulary; Porter stemming only as a lower-confidence fallback for unknown English terms; and exact original-text matching as a permanent fallback. Search results should show the reason for a match (exact, inflectional, derivational relation or stem fallback).

The design should not conflate clinically distinct concepts automatically. For example, infection and inflammation must remain distinct. A clinician should see the original wording, normalized representation and any uncertainty flag. The system must not rewrite source records.

11.3 Expected failures and clinician-visible behaviour

- Unknown drug brand or misspelling: display as unanalysed text and preserve exact search.
- Ambiguous abbreviation: show the original abbreviation and do not force one expansion.
- Incorrect OCR character: present a suggested candidate with uncertainty rather than silently correcting.
- Missing lexicon entry: fall back to character n-gram or stem retrieval, clearly labelled as approximate.
- Negation and temporality: do not treat “no fracture” or “history of fracture” as a current confirmed diagnosis.

11.4 Effort and cost

Manual work includes pattern writing, clinical lexicon curation, affix review, annotation of 200 word forms, and relevance judgements for 40 queries. The initial finite-state layer has a higher setup cost but is predictable and explainable. Statistical methods become more economical when the vocabulary changes rapidly, there are many regional-language variants, or the number of manually curated exceptions becomes too large. Even then, deterministic validators should remain for high-value structured fields.

11.5 Decision

The hybrid design is preferred over maximum recall alone because the cost of a misleading clinical match can be greater than the cost of missing a weakly related result. Use precision thresholds and human review for approximate matches. The system is a search aid and must not be used to infer a diagnosis or treatment.

TASK 12 – DAY-4 DESIGN CHALLENGE: OCR AND CODE-MIXED INPUT

12.A Impact analysis

The new inputs violate four original assumptions: clean typed characters, stable word boundaries, a closed English vocabulary and exact matching. OCR introduces substitutions such as 0/O, 1/l and rn/m; line breaks may split words; diacritics may disappear. Romanised regional language mixed with English introduces non-English morphology and code-switching at token level. The brief expects at least 20% of incoming tokens to be out-of-lexicon, misspelt or non-English.

New input property	Original component affected	Likely consequence
0/O, 1/l, rn/m confusion	Regex and DFA	False negatives or wrong identifiers
Line-end hyphenation	Tokenizer	One word split into two tokens
Lost diacritics	Lexicon FST	Unknown or conflated forms
Romanised regional language	English lexicon	Low analyser coverage
Token-level code-mixing	Language assumptions	Incorrect expansion and stemming

A 20% noisy/OOV token rate does not imply exactly 20% retrieval loss: some noisy tokens may be irrelevant, while one corrupted identifier can prevent an otherwise correct record match. Precision and recall must be measured separately for each input type.

12.B Original design on three new records

Record type	Example input	First failure in original pipeline	Reason
OCR	PT-12O456; fractu re noted	Identifier DFA / tokenizer	Letter O replaces zero; line break splits word
Code-mixed	patient ku fever irundhuchu, paracetamol 500mg kuduthen	English morphological analyser	Regional-language tokens are out of lexicon
Both	PT-12O456; patient ku fev er; 500 mg paracetamol	Entity and morphology stages	OCR corruption plus code-mixing

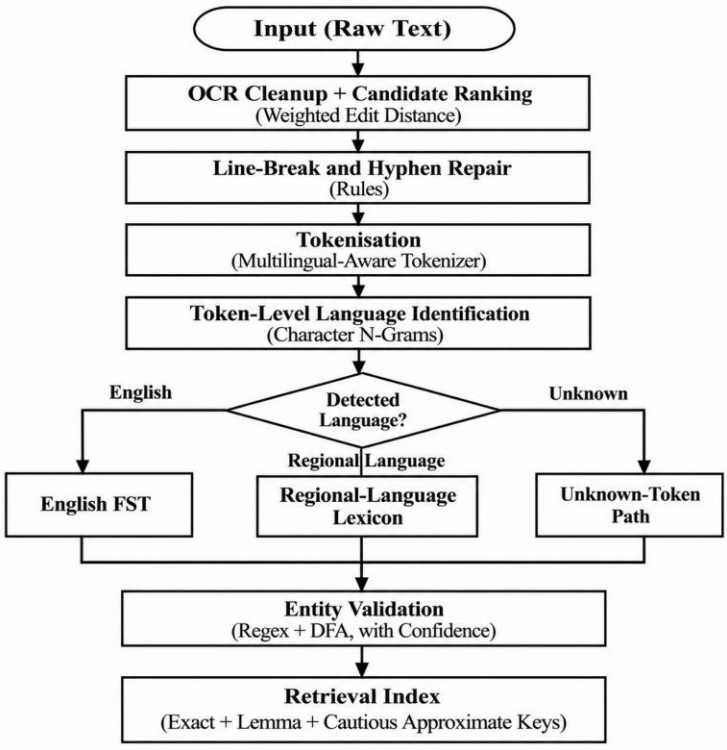
These are hand-traced test cases, not measured evaluation records. The original exact-match design rejects the corrupted identifier and leaves regional-language words unanalyzed. The drug-strength entity may still be recognized because “paracetamol 500mg” is English and numeric.

12.C Evaluate candidate formalisms

Method	Use	Strength	Risk/limitation
Weighted finite-state transducer	Rank OCR correction candidates	Combines edit cost with lexical knowledge	May choose a plausible but wrong clinical term
Edit-distance matching	Recover near-miss words/IDs	Simple and language-independent	Needs thresholds; can create false matches
Character n-gram language ID	Identify language per token	Useful for code-mixed text	Short tokens and shared vocabulary are difficult
Regional-language lexicon/FST	Analyse known Romanised forms	Interpretable for covered vocabulary	Requires expert curation and variant maintenance

12.D Adaptation: modify, do not replace

The recommended decision is MODIFY. Retain exact regex/DFA validation for clean structured fields, but place a cautious OCR correction layer before entity recognition. Add line-break repair, language identification and regional-language resources before morphological analysis. Preserve both the raw OCR string and corrected candidate.



12.E Exact component changes

Component	Change	Expected effect
Patterns	Allow controlled whitespace/hyphen variants; keep strict ID	Higher recall; possible extra candidates

	checksum/format checks	
Automata	Add correction candidate input; validate corrected value separately	More OCR recovery without weakening final validation
Lexicon	Add regional stems, transliteration variants and common misspellings	Higher coverage; more annotation effort
Rule cascade	Add line-wrap repair and diacritic normalisation before morphology	Fewer split/unknown tokens
Retrieval	Separate exact, lemma, transliteration and approximate scores	Better recall with transparent ranking

12.F Evidence-based defence and limitations

The modification follows from the changed input conditions: exact automata remain valuable for validating clean strings, but exact matching alone is too brittle for OCR. A weighted correction layer can propose alternatives; a language identifier routes tokens to the correct lexicon; and the original text remains visible. The design should be evaluated on a labelled noisy sample and compared against the clean baseline.

Illustrative target evaluation (not measured): analyser coverage may fall from 75% on clean English to 55–60% on mixed noisy input without adaptation; the modified pipeline may recover coverage to 68–72%, while precision may fall slightly due to correction candidates. These ranges are planning hypotheses only. Report actual precision, recall, F1 and coverage after annotating the noisy sample.

A finite-state layer will never fully solve unrestricted semantics, deep discourse, clinical negation scope in every construction, novel slang, arbitrary OCR ambiguity or unseen language varieties without additional resources. Human review remains necessary for uncertain corrections and clinically consequential interpretation.

CONCLUSION

This assignment develops a layered approach to morphology-aware clinical text normalisation and search. Regular expressions identify candidate structured entities; finite automata validate exact formats; tokenisation and rewrite rules standardise surface text; finite-state morphology connects surface forms to lemmas and grammatical features; and Porter stemming provides a broad but less precise fallback.

The central design principle is to preserve original text and make every transformation traceable. Clinical search requires a balance between recall and precision. Aggressive stemming or abbreviation expansion can create misleading matches, so the system should expose match type and uncertainty. Under OCR and code-mixed conditions, the finite-state layer should be supplemented with weighted correction, language identification and additional lexical resources.

REFERENCES

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APPENDIX A – SAMPLE SYNTHETIC CLINICAL RECORDS

Record SYN-0001 | Discharge summary | Patient PT-100001, 45 years, admitted 04/11/2026 with fever for 3 days. Temperature 37.8 °C. Diagnosis recorded as urinary tract infection. Discharged 06/11/2026.

Record SYN-0002 | Prescription | Patient PT-100002. Paracetamol 500 mg PO TDS for 3 days. Review if symptoms persist. No known allergy documented.

Record SYN-0003 | Operation note | Patient PT-100003 underwent appendectomy. Ward B2, Bed 14. BP 120/80 mmHg. Procedure completed without documented complication.

Record SYN-0004 | Nursing observation | Patient PT-100004 reports pain for 6 hours. HR 88 bpm, RR 18/min, temperature 37.2 °C. Continue monitoring.

These four records illustrate the four record types only. They do not satisfy the 500-record requirement. Generate at least 496 additional synthetic records using varied templates, dates, units, abbreviations, spelling variants and clinical vocabulary, then document the generation procedure.

APPENDIX B – QUERY AND ANNOTATION DESIGN

Each clinician query should include query text, intended clinical concept, relevant record IDs, non-relevant sampled record IDs, and assessor initials or anonymised assessor code. Two reviewers should judge relevance independently where possible; disagreements should be resolved and documented.

Query ID	Example query	Relevant target	Judgement fields
Q01	fracture	fracture, fractures, fractured, fracturing records	Relevant / not relevant
Q02	renal surgery	nephrectomy and related procedure records	Relevant / not relevant
Q03	paracetamol 500 mg	matching prescription records	Relevant / not relevant
Q04	high blood pressure	hypertension and BP observations	Relevant / not relevant

For the 200 word-form annotation set, record surface form, lemma, part of speech, inflectional features, derivational structure, ambiguity notes and annotator decision. Include regular and irregular forms, abbreviations, misspellings and code-mixed examples.

APPENDIX C – METRICS AND REPORTING FORMULAE

- Precision = TP / (TP + FP). It measures the proportion of retrieved items that are relevant.
- Recall = TP / (TP + FN). It measures the proportion of relevant items retrieved.
- F1 = 2 × (Precision × Recall) / (Precision + Recall). It is the harmonic mean of precision and recall.
- Coverage = number of tokens receiving a valid analysis / total eligible tokens.
- Type-to-lemma ratio = distinct surface types / distinct lemmas.
- Runtime = elapsed wall-clock time for the specified stage, measured on the same machine and repeated runs.
- Effort = person-hours spent on corpus generation, lexicon curation, rule writing, annotation, testing and maintenance.

Always state the denominator and whether the result is micro-averaged, macro-averaged or per-query. Do not compare metrics calculated using different relevance definitions.

APPENDIX D – SUBMISSION CHECKLIST

- Replace student details and submission date on the cover page.
- Build and document at least 500 de-identified or synthetic records and at least 25,000 tokens.
- Expand the starter lexicon to at least 300 stems, 20 inflectional affixes and 15 derivational affixes.
- Annotate at least 200 word forms and prepare at least 40 clinician queries with relevance judgements.
- Run all patterns and automata; replace illustrative counts with actual TP, FP and FN.
- Execute the code and record actual runtime, coverage, precision, recall and F1.
- Generate at least one plot from measured data and include the underlying table.
- Verify every medication, unit and clinical example with an appropriate domain reviewer.
- Keep original text and mark all approximate or uncertain matches.
- Proofread page numbers, headings, tables and references before submission.

SUPPLEMENTARY PAGE – REGULAR EXPRESSION TEST CASES

The following test cases extend the entity pattern suite. Each test should be run against the exact production pattern and manually reviewed. A syntactic match is only a candidate; semantic validation remains a separate step.

Entity	Input	Expected
Patient ID	PT-654321	Accept
Patient ID	PT-65432	Reject
Date	29/02/2024	Format accept; calendar accept
Date	29/02/2025	Format accept; calendar reject
Drug strength	amoxicillin 250 mg	Candidate accept
Drug strength	250 mg amoxicillin	Reject for this pattern

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – AUTOMATON TESTING NOTES

A complete DFA test suite must include accepted strings, rejected strings, empty input, truncated input, extra trailing characters and characters outside the alphabet. A dead state makes rejection explicit and prevents accidental partial acceptance.

Test	Input	Expected result
Valid identifier	PT-123456	Accept
Short identifier	PT-12345	Reject
Extra character	PT-123456X	Reject
Wrong prefix	PX-123456	Reject
Empty string		Reject

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – LEXICON EXPANSION PLAN

The minimum 300-stem lexicon should be built from documented sources and reviewed. A proposed distribution is 80 nouns, 70 verbs, 60 adjectives, 50 clinical procedure/condition terms and 40 abbreviations or lexicalised forms. Do not count inflected variants as separate stems.

Category	Target stems	Examples
Anatomy	60	kidney, heart, liver
Conditions	70	fracture, infection, fibrosis

Procedures	50	biopsy, nephrectomy
Clinical actions	60	monitor, admit, discharge
Descriptors	60	acute, chronic, stable

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – IRREGULAR MORPHOLOGY RESOURCE

Irregular forms must be stored as explicit mappings with features. The lexicon should also record whether a form is a noun, verb or adjective, because the same surface string may have multiple analyses.

Surface	Lemma	Feature
was	be	V; past; singular
were	be	V; past; plural
had	have	V; past
has	have	V; 3SG
children	child	N; plural
teeth	tooth	N; plural
feet	foot	N; plural

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – CLINICAL ABBREVIATION SAFETY

Abbreviation expansion is context-sensitive. The system should maintain an abbreviation table with allowed meanings, context cues, source and reviewer. Unresolved abbreviations remain unchanged and are indexed in their original form.

Abbreviation	Possible meaning	Handling
OD	Once daily / other meanings	Context required
MS	Multiple sclerosis / other meanings	Do not expand globally
RA	Rheumatoid arthritis / other meanings	Context required
BP	Blood pressure	Usually safe in vitals context
PRN	As needed	Expand when prescription context supports it

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – OCR CORRECTION POLICY

OCR correction should be candidate generation, not silent replacement. A corrected token must retain a link to the original token, confidence score, correction rule and alternatives. Structured identifiers should be verified through an authoritative source or checksum where available.

Observed OCR	Candidate	Caution
PT-12O456	PT-120456	O/0 ambiguity
fractu re	fracture	Whitespace repair
rnass	mass	rn/m confusion
l0 mg	10 mg	l/1 ambiguity

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – CODE-MIXED LANGUAGE HANDLING

Romanised regional-language text is not necessarily misspelt English. Token-level language identification should route tokens to the relevant language resource. For mixed phrases, preserve token order and do not translate or expand a regional token unless a validated lexicon supports the mapping.

Example	Observation	Recommended action
patient ku fever	English + regional case marker + English	Keep mixed analysis
fever irundhuchu	English noun + regional verb	Route token by token
paracetamol 500mg	Drug + strength	Use English entity pattern
kuduthen	Regional-language verb form	Leave OOV if no lexicon

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – ERROR TAXONOMY

Every failure should be assigned a reproducible category. This supports targeted improvements and prevents the team from treating all failures as generic NLP errors.

Error code	Description	Example
E01	Boundary/tokenisation	fractu re
E02	Character/OCR	PT-12O456
E03	Unknown lexeme	unlisted drug brand
E04	Ambiguous abbreviation	MS

E05	Morphology rule gap	irregular plural
E06	Semantic mismatch	inflammation vs infection

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – RETRIEVAL EVALUATION PROCEDURE

Run the same 40 queries against the same frozen record collection under each strategy. Use identical relevance judgements and record retrieved IDs, rank positions and assessor decisions. Do not change the corpus between strategies.

Run	Configuration	Record
A	Exact search only	Baseline
B	Exact + Porter	Stem fallback
C	Exact + FST	Lemma and features
D	Hybrid with noisy-input support	Separate approximate channel

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – RISK AND GOVERNANCE

Because the corpus contains clinical text, governance is essential even for a student prototype. Synthetic data is preferred. If de-identified records are used, institutional approval and applicable privacy requirements must be followed. The tool must not expose patient identity through logs or exported reports.

Risk	Control
Re-identification	Use synthetic IDs; remove direct identifiers
Wrong abbreviation expansion	Retain original and ambiguity flag
Incorrect OCR correction	Show alternatives and confidence
Over-stemming	Label approximate match; preserve exact text
Unauthorized access	Role-based access and audit logging

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – REPRODUCIBILITY

A reproducible experiment records corpus version, random seed for synthetic generation, software versions, hardware, configuration, query set, annotation guidelines and code commit. Save raw output so tables can be regenerated.

Item	Record
Python	Version used
NLTK	Version used
Corpus	Version/hash
Hardware	CPU/RAM/OS
Random seed	Fixed integer
Run count	At least 3 repeated runs

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – SAMPLE QUERY SET EXTENSION

The following examples can be extended to forty queries. Each query should have a clear target concept and relevance rules. Avoid judging a record relevant merely because it contains a shared substring.

ID	Query	Target
Q05	fractured femur	Femur fracture forms
Q06	renal removal	Nephrectomy records
Q07	temperature 37.5 C	Temperature observations
Q08	blood pressure 120/80	Matching BP measurements
Q09	three times daily	TDS frequency
Q10	post-operative review	Postoperative notes

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – SAMPLE ANNOTATION FORMAT

Annotation must distinguish lemma from stemmer output. It must also record grammatical features and uncertainty. Use a stable tag set across annotators.

Surface	Lemma	POS	Features	Confidence
fractures	fracture	N	PL	High
treated	treat	V	PAST	High
drains	drain	N/V	PL or 3SG	Ambiguous
children	child	N	PL	High
nephrectomy	nephrectomy	N	DERIVED	High

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – DESIGN DECISION MATRIX

A design matrix makes the recommendation explicit. Weight clinical precision and traceability more heavily than raw recall because the system supports clinical record search.

Criterion	Exact	Porter	FST	Hybrid
Precision	High	Medium	High	High
Recall	Low	Medium-high	High in-vocab	High
OOV handling	Low	Medium	Low	Medium-high
Explainability	High	Medium	High	High
Maintenance	Low	Low	High	High

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – IMPLEMENTATION LIMITATIONS

The compact code is a reference skeleton. It does not yet implement all required 12 entity patterns, a complete NFA/DFA minimizer, a full 300-stem FST, all orthographic transducers or an end-to-end benchmark. These are explicit remaining implementation tasks rather than claims of completed execution.

Requirement	Status in this report	Action
12+ patterns	Representative patterns supplied	Implement and test all
4 automata	Designs described	Complete transition tables/code
300 stems	Resource plan supplied	Populate lexicon
200 annotations	Format supplied	Annotate forms
40 queries	Examples supplied	Complete relevance set
Runtime/metrics	Illustrative only	Run experiment

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

SUPPLEMENTARY PAGE – FINAL CLINICAL SAFETY CHECK

Before deployment, validate date conventions, medication units, abbreviation meanings, negation, temporality and patient identifiers with domain experts. A search result is a pointer to source documentation, not a clinical conclusion. The user interface should clearly distinguish exact matches from approximate matches.

Check	Expected safeguard
Medication	Never alter dose or unit silently
Negation	Preserve “no”, “without”, “rule out”
Temporality	Distinguish past history from current condition
Dates	Flag ambiguous formats
Identity	Use approved identifier validation
Approximate search	Show reason and confidence

The examples on this page are design and test specifications. They should be converted into executable tests and measured results when the final corpus and implementation are available.

DETAILED DISCUSSION – WHY PRESERVE THE ORIGINAL TEXT

Normalisation is a derived view of the record. The source document may contain clinically meaningful punctuation, capitalisation, shorthand, uncertainty or a correction. If the pipeline overwrites the original, it becomes impossible to audit why a record was retrieved or to recover information lost during case folding, abbreviation expansion or OCR repair. Store raw text, normalized text, token offsets and transformation provenance together.

Design implication: the implementation should expose this decision in its configuration and test suite. The report should state whether the behaviour was measured, manually traced or proposed, and should never present an unexecuted experiment as completed evidence.

For the hospital setting, auditability is as important as raw retrieval performance. Retain original spans, record every rewrite, and let the clinician inspect the source record before drawing conclusions.

DETAILED DISCUSSION – OFFSET PRESERVATION

Every token should carry its start and end character offsets in the original document. If a normalized token expands from “BP” to “blood pressure”, the output should retain a pointer to the original two-character span. Offset preservation allows the interface to highlight the exact source phrase and helps reviewers verify an extraction.

Design implication: the implementation should expose this decision in its configuration and test suite. The report should state whether the behaviour was measured, manually traced or proposed, and should never present an unexecuted experiment as completed evidence.

For the hospital setting, auditability is as important as raw retrieval performance. Retain original spans, record every rewrite, and let the clinician inspect the source record before drawing conclusions.

DETAILED DISCUSSION – CLINICAL NEGATION

The phrase “no evidence of fracture” contains the word fracture but does not assert that a fracture is present. Similarly, “rule out infection” and “history of renal disease” have different certainty and time status. A morphology-aware layer does not solve negation or temporality. A later context module should represent assertion, experiencer, temporality and certainty, and the user interface should avoid presenting keyword matches as confirmed diagnoses.

Design implication: the implementation should expose this decision in its configuration and test suite. The report should state whether the behaviour was measured, manually traced or proposed, and should never present an unexecuted experiment as completed evidence.

For the hospital setting, auditability is as important as raw retrieval performance. Retain original spans, record every rewrite, and let the clinician inspect the source record before drawing conclusions.

DETAILED DISCUSSION – DATE VALIDATION

A regular expression can validate a date's shape but cannot reliably establish the calendar validity of every date. After extracting a candidate, use a date parser with a known locale. For day/month/year input, validate leap years and month lengths. Ambiguous values should remain unresolved until the source-system convention is known.

Design implication: the implementation should expose this decision in its configuration and test suite. The report should state whether the behaviour was measured, manually traced or proposed, and should never present an unexecuted experiment as completed evidence.

For the hospital setting, auditability is as important as raw retrieval performance. Retain original spans, record every rewrite, and let the clinician inspect the source record before drawing conclusions.

DETAILED DISCUSSION – UNIT NORMALISATION

Unit normalization should separate lexical standardization from numeric conversion. “500 milligrams” and “500 mg” can share a canonical unit label. Converting 0.5 g to 500 mg requires a numeric conversion with dimensional checking. The system should preserve the original quantity and unit and log the conversion.

Design implication: the implementation should expose this decision in its configuration and test suite. The report should state whether the behaviour was measured, manually traced or proposed, and should never present an unexecuted experiment as completed evidence.

For the hospital setting, auditability is as important as raw retrieval performance. Retain original spans, record every rewrite, and let the clinician inspect the source record before drawing conclusions.

DETAILED DISCUSSION – SEARCH INDEX DESIGN

An inverted index maps each normalized term or feature to the records in which it appears. Maintain separate fields for exact tokens, lemmas, structured entities, abbreviation expansions

and approximate OCR candidates. Search ranking can then prioritize exact matches, followed by lemma matches, then carefully weighted approximate matches.

Design implication: the implementation should expose this decision in its configuration and test suite. The report should state whether the behaviour was measured, manually traced or proposed, and should never present an unexecuted experiment as completed evidence.

For the hospital setting, auditability is as important as raw retrieval performance. Retain original spans, record every rewrite, and let the clinician inspect the source record before drawing conclusions.

DETAILED DISCUSSION – CLINICIAN QUERY JUDGEMENTS

A query relevance judgement should be based on the clinical intent, not merely string overlap. For “renal surgery”, a nephrectomy record may be relevant even when the word renal is absent. However, a record mentioning renal function tests may not be relevant to a surgery query. The annotation guideline should define inclusion and exclusion examples.

Design implication: the implementation should expose this decision in its configuration and test suite. The report should state whether the behaviour was measured, manually traced or proposed, and should never present an unexecuted experiment as completed evidence.

For the hospital setting, auditability is as important as raw retrieval performance. Retain original spans, record every rewrite, and let the clinician inspect the source record before drawing conclusions.

DETAILED DISCUSSION – PRECISION AND RECALL TRADE-OFF

Increasing normalization generally improves recall by connecting variants, but it can reduce precision when unrelated terms are conflated. In clinical retrieval, a high-recall candidate list can be useful if results are clearly ranked and reviewed, but false equivalence must not be hidden. Evaluate both measures and inspect the error types.

Design implication: the implementation should expose this decision in its configuration and test suite. The report should state whether the behaviour was measured, manually traced or proposed, and should never present an unexecuted experiment as completed evidence.

For the hospital setting, auditability is as important as raw retrieval performance. Retain original spans, record every rewrite, and let the clinician inspect the source record before drawing conclusions.

DETAILED DISCUSSION – FINITE-STATE RESOURCE MAINTENANCE

Finite-state resources are maintainable when updates are versioned and reviewed. A new drug or procedure term should be added with its part of speech, known variants, derivational family and provenance. Rule changes should be accompanied by regression tests so that a new spelling rule does not break existing forms.

Design implication: the implementation should expose this decision in its configuration and test suite. The report should state whether the behaviour was measured, manually traced or proposed, and should never present an unexecuted experiment as completed evidence.

For the hospital setting, auditability is as important as raw retrieval performance. Retain original spans, record every rewrite, and let the clinician inspect the source record before drawing conclusions.

DETAILED DISCUSSION – CONCLUSION ON DAY-4 CHALLENGE

The changed input does not make finite-state methods obsolete. It changes their position in the pipeline. Exact finite-state recognition remains valuable after OCR candidate generation and language routing. A hybrid architecture preserves deterministic validation while adding uncertainty-aware handling for noisy and mixed-language tokens.

Design implication: the implementation should expose this decision in its configuration and test suite. The report should state whether the behaviour was measured, manually traced or proposed, and should never present an unexecuted experiment as completed evidence.

For the hospital setting, auditability is as important as raw retrieval performance. Retain original spans, record every rewrite, and let the clinician inspect the source record before drawing conclusions.